

Detection of Polycystic Ovary Syndrome (PCOS) by Transformer-Vision and Deep Learning Approach

by

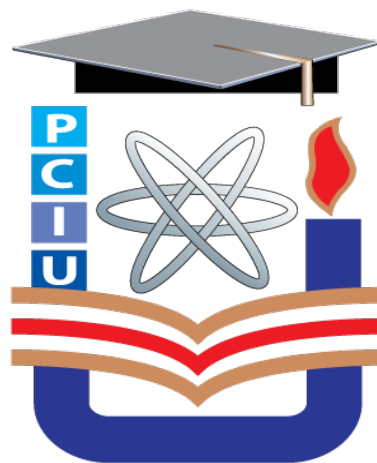
Mohammad Imam Uddin
CSE 01906822

Kazi Md. Jonayed
CSE 01606607

PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND ENGINEERING

Under the supervision of

Mr. Rayhanuzzaman
Lecturer



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
PORT CITY INTERNATIONAL UNIVERSITY

May 2024

Declaration of Originality

Perhaps it's best to state that to the best of the author's knowledge, this thesis has not been submitted anywhere for the award of any degree. References and citations to other sources have been included as appropriate. We also certify that we only used the materials that were mentioned. All formulations and notions taken directly or in substance from printed or unprinted literature, or the Internet, have been properly cited with footnotes or other exact references to the original source, as per the standard scientific procedure. We are mindful that disseminating foundational information could have significant consequences.



(Signature of the candidate)

Mohammad Imam Uddin

CSE 01906822

(Signature of the candidate)

Kazi Md. Jonayed

CSE 01606607

Approval for Submission

This is to certify that the thesis entitled **Detection of Polycystic Ovary Syndrome (PCOS) by Transformer-Vision and Deep Learning Approach** submitted by **Mohammad Imam Uddin** (CSE 01906822) and **Kazi Md. Jonayed** (CSE 01606607) are undergraduate students of the **Department of Computer Science and Engineering** has been examined and evaluated. I hereby accord my approval of this thesis as the presented work is deemed satisfactory and meets the requirements for submission.

Place: Chattogram

Date: May, 2024

Rayhanuzzaman

(Signature of Supervisor)

Mr. Rayhanuzzaman

Lecturer

Department of Computer Science and Engineering

Port City International University

7, Nikunja Housing Society, South Khushi, Chattogram- 4202.

Chittagong, Bangladesh.

Email: rayhanuzzamanr@gmail.com

Telephone: 01722600566

Dedication

Dedicated to our cherished parents and our respectable teachers.

Table of Contents

Declaration	i
Approval	ii
Table of Contents	iv
List of Figures	viii
List of Tables	ix
Acknowledgments	x
Abstract	xi
Chapter 1 Introduction	1
1.1 Background Information	1
1.1.1 Purpose of study	2
1.1.2 Objectives	2
1.2 Problem Statement	2
1.2.1 Significance	3
1.2.2 Motivation	4
1.3 Problem Definition	4
1.4 Assumptions	4
1.5 Limitations	5
1.6 Delineation	6
1.7 Research questions	6
1.8 Brief chapter overviews	7
Chapter 2 Literature Review and Analysis	8
2.1 Introduction	8
2.2 Related work in PCOS detection	8
2.3 Conclusion	11
Chapter 3 Methodology	12
3.1 Introduction	12
3.2 Research Design	12
3.3 Research instruments	13

3.3.1	Python	13
3.3.2	Colab	14
3.3.3	Hardware implementation	14
3.3.4	Datasets	15
3.3.5	Data pre-processing	15
3.3.6	Feature extraction	17
3.3.7	Splitting into training and testing	17
3.4	Random over-sampling	17
3.5	Analysis	18
3.5.1	CNN	18
3.5.2	Inception V3	20
3.5.3	The Transformer: Attention is all you need	20
3.5.4	Vision Transformers	24
3.5.5	Hyperparameter tuning	28
3.5.6	Limitations	29
3.5.7	Ethics	29
3.6	Conclusion	30
Chapter 4	Finding and Analysis	31
4.1	Precision, Recall and F1-scores	31
4.1.1	CNN Classification Report	31
4.1.2	Inception V3 Classification Report	32
4.1.3	ViT-Base Classification Report	32
4.1.4	DeiT Classification Report	33
4.1.5	BeiT Classification Report	33
4.1.6	ViT-Large Classification Report	34
4.1.7	Conclusion	34
4.2	Confusion Matrix	34
4.2.1	Conclusion	37
4.3	Train-Validation loss curves	38
4.3.1	Conclusion	40
4.4	Comparative Analysis	41
4.5	Experimental Input and Output	41

Chapter 5	Conclusion	42
5.1	Summery of findings	42
5.2	Conclusion	42
5.3	Summery of contribution	42
5.4	Future work	43
References		44

List of Figures

3.1	Steps in Developing the PCOS Detection Model	12
3.2	Dataset Vsualization	15
3.3	Edge Detection	16
3.4	Binarization	16
3.5	ORB	16
3.6	CNN Architecture	19
3.7	Convolution layer	19
3.8	Inception V3 architecture	21
3.9	Single and multi-headed self-attention architecture	22
3.10	Architecture of the Vision Transformer	24
3.11	Attention maps. ^[30]	25
3.12	Transfer of large datasets on large models	26
3.13	ViT vs CNN Pre-training Dataset Size Comparison	27
3.14	DeiT Architecture	28
3.15	BEiT Architecture	29
4.1	CNN Classification Report	31
4.2	Inception V3 Classification Report	32
4.3	ViT-Base Classification Report	32
4.4	DeiT Classification Report	33
4.5	BEiT Classification Report	33
4.6	ViT-Large Classification Report	34
4.7	CNN Confusion Matrix	35
4.8	Inception v3 Confusion Matrix	35
4.9	ViT Base Confusion Matrix	36
4.10	DeiT Confusion Matrix	36
4.11	BEiT Confusion Matrix	37
4.12	ViT Large Confusion Matrix	37
4.13	CNN Accuracy and Loss Curve	38
4.14	Inception V3 Accuracy and Loss	38
4.15	ViT Base Accuracy and Loss	39

4.16 DeiT Accuracy and Loss	39
4.17 BeiT Accuracy and Loss	40
4.18 ViT Large Accuracy and Loss	40
4.19 Final Outcome	41

List of Tables

3.1	Splitting into training and testing	17
4.1	Comparative Analysis	41
4.2	Performance Metrics for PCOS Detection	41

Acknowledgments

Above all, we thank Allah from the bottom of our hearts for giving us a wonderful opportunity to successfully complete our final year thesis. We had the incredible privilege of concentrating under the guidance and support of Mr. Raihanuzzaman. His breadth of knowledge gives us the opportunity to broaden our horizons and make meaningful progress. We would also want to express our gratitude to the Department of Computer Science and Engineering for providing us with the resources necessary to execute the program. We finally want to express our gratitude to our family for their assistance and encouragement during our evaluations.

Abstract

Polycystic ovary syndrome is one of the most common endocrinological disorders affecting many women of reproductive age around the world. PCOS is characterized by a range of symptoms, including acne, hair loss, and hyperandrogenemia. It is also one of the leading causes of female infertility. It is essential to detect PCOS early on due to its detrimental health effects. In this thesis, we present a novel approach for PCOS detection using transformer-vision and deep learning. We have used ViT-based models, ViT Base and ViT Large, to analyze ovary ultrasound images for PCOS detection. Our results show remarkable accuracy, with ViT Base reaching 98.88% accuracy and ViT Large reaching 99.18%. We also compare our proposed transformer-vision based models with previously developed Convolutional neural network (CNN) models as well as PCONet and a fine-tuned idea InceptionV3, confirming the superiority of our transformer-vision models for PCOS detection from ultrasound images.

Keywords: Polycystic Ovary Syndrome, Transformer-Vision, Deep Learning, ViT Base, ViT Large, PCOS ultrasound image classification, Polycystic ovary detection, PCOS detection.

Chapter 1

Introduction

1.1 Background Information

Polycystic ovary syndrome (PCOS) is recognized as the most common endocrine disorder among women of reproductive age worldwide.^[1] Since polycystic ovarian disease (PCOD) was first recognized, Michael Leventhal and Irving Stein ^[2] introduced the triple terms "obesity," "hirsutism," and "amenorrhea". In 1935, their relationship was discovered between birth defects and obesity, hence also called 'hyperandrogenic anovulation' (HA) or 'Stein-Leventhal syndrome'.^[3] It is the most common health problem affecting women in their reproductive years, 5 to 10% of them have PCOS ^[4]. It is also known as "Syndrome O", which refers to nutrient overload, insulin overproduction, fetal hypochondriasis, and ovarian dysfunction. Therefore, PCOD is also known as polycystic ovary syndrome (PCOS). Infertility is defined as the inability of the ovaries to release the egg. When the follicles grow abnormally big and in large amounts during the ovulatory stage, it is considered the first symptom of Polycystic Ovary Syndrome (PCOS).^[5] However, there are many other reasons for infertility. Patients with a familial history of PCOS had a 24% to 32% increased risk of PCOS, but untreated premenopausal women were more likely to develop PCOS.^[6]

Ultrasound is the most commonly used imaging modality in the clinical evaluation of patients with endometriosis. Ultrasound has many advantages over other medical imaging modalities such as CT and MRI. One of the main advantages of ultrasound is its low cost, ease of use and safety. It also offers real-time diagnostic results. The imaging technique of ultrasound also offers a great opportunity for the development of deep learning models for automatic analysis. This makes the examination more impartial and accurate in diagnosing the disease. Deep learning is one of the most powerful techniques used in imaging analysis as well as computer vision. Automatic classifying of PCOS at the early stage based on ultrasound images and clinical information helps in disease recognition.^[7]

It is important to detect PCOS as soon as possible in order to begin treatment. Machine learning can be used as an alternative to traditional methods of diagnosing PCOS, which are often time-consuming.

1.1.1 Purpose of study

The purpose of the study is to explore and develop novel techniques for the diagnosis of polycystic ovary syndrome (PCOS), based on the principles of **Vision Transformers** and Deep Learning. The goal of the research is to improve the identification of PCOS from ultrasound pictures with the help of advanced computational techniques such as ViT models. The aim is to advance medical imaging and the field of reproductive health by helping to detect and treat polycystic ovaries early.

1.1.2 Objectives

- Investigate how doctors identify PCOS on ultrasound image scans.
- Evaluate the issues with commonly used PCOS detection techniques.
- Check how efficient CNNs can identify PCOS from ultrasonogram.
- Examine whether transformer-based models, like ViT, can improve the efficiency and accuracy of Polycystic ovary syndrome (PCOS) detection.
- Develop novel approaches to PCOS diagnosis with transformer vision and deep learning.
- Compare the detection of PCOS using transformer-based models with standard CNN approaches.
- To ensure the new techniques are effective, test them.
- Provide suggestions on how to use clever computer approaches in the future to improve PCOS identification.

1.2 Problem Statement

Polycystic ovary syndrome (PCOS) is a commonplace endocrine disease affecting girls of reproductive age, characterized with the aid of hormonal imbalance, ovarian dysfunction. Early diagnosis of PCOS is critical for timely intervention and management deal with the related health risks consisting of infertility and but modern-day PCOS diagnostic strategies, particularly ultrasound- Imaging-primarily based, frequently face challenges of accuracy, performance and interpretation of outcomes around

Traditional techniques of PCOS diagnosis depend upon manual exam of ultrasound pictures through healthcare specialists, which may be subjective and time-ingesting. In addition, existing pc-assisted strategies, along with convolutional neural networks (CNNs), despite the fact that promising, can not completely cope with the complexity and nuances of PCOS prognosis

In addition, the arrival of transformer-based totally fashions inclusive of the Vision Transformer (ViT) affords an possibility to enhance PCOS detection via advanced photograph processing strategies but enhancing these fashions to improve the accuracy and efficiency of PCOS analysis is still unclear.

Thus, in order for PCOS screening bifurcations to be more accurate, efficient, and accessible, the shortcomings of state-of-the-art methods for PCOS screening must be addressed and alternatives should be explored a transformer-based total designs are adopted. Its potential is tested, and new strategies are proposed to address current challenges in the diagnosis of PCOS.

1.2.1 Significance

The importance of this have a look at, which highlights the important position of new methods in the diagnosis of PCOS, may be outlined as follows:

- **Better health care:** Through the use of more advanced digital technologies, accurate and environmentally friendly analysis of PCOS can lead to early detection and intervention, ultimately creating a health haven for women who suffer from this condition.
- **Better reproductive health:** Early detection of PCOS allows for early intervention and management, which reduces infertility and other issues related to this condition.
- **Personalized health care:** People with proven PCOS can improve treatment by having access to accurate and reliable diagnostic tools, which reflect their right to treatment a personalized program that is almost entirely tailored to their needs.
- **Research Breakthrough:** Transformer-Based Application of New Techniques for PCOS Diagnosis General trends contribute to the advancement of computational and clinical imaging techniques, opening the door for future reproductive endoscopy.

- Public health implications: Because PCOS affects the majority of young women worldwide, addressing complex prognostic conditions has important public health implications. Advanced diagnostic methods can better identify problems, provide solutions, and provide preventive measures to prospective audiences.

1.2.2 Motivation

In the face of an urgent need to revolutionize PCOS diagnosis for women worldwide, and raise standards of health care, a strong commitment to using advanced digital technologies and transformer-based approaches role to overcome significant challenges, with a strong focus on pioneering new frontiers in innovation and motivating this review , this review aims to significantly improve the diagnosis of PCOS and ultimately to aims to achieve tangible improvements in the lives and health of women affected by the condition around the world.

1.3 Problem Definition

The issue we aim to address concerns the shortcomings and difficulties of the present approaches to diagnosing polycystic ovarian syndrome (PCOS), particularly in the area of clinical imaging. The accuracy, efficiency, and scalability issues with current approaches often result in prolonged pathology and unfavorable consequences. Additionally, the intensity of PCOS symptoms makes it more difficult to examine the clinical picture and diagnose such serious disorders, which hinders adequate diagnosis and treatment management.

In addition, the absence of a standardized diagnostic method makes clinical practice challenging and imprecise, with results dependent on subjective interpretation. This makes medical imaging much more difficult, increasing the possibility of errors and inconsistent diagnoses and necessitating the use of precise, effective, and accurate PCOS diagnostic techniques.

1.4 Assumptions

1. The concept that medical imaging plays an important role in the diagnosis of PCOS and advances in imaging technology may lead to more accurate diagnosis.
2. The concept that PCOS symptoms can be reliably captured and interpreted using medical imaging techniques, despite the complexity and inherent variability of the condition.

3. The idea is that machine learning algorithms can classify and categorize PCOS-related information in medical documentation and learn more effectively from clinical data sets.
4. The concept that standardized diagnostic criteria for PCOS exist and can be effectively applied in health care settings.
5. The assumption that the availability of annotated medical data to train machine learning models is sufficient to achieve high accuracy in PCOS diagnosis.
6. The concept that the use of alternative diagnostic methods for PCOS will be welcomed and adopted by health professionals and organizations.
7. It is hypothesized that using transformer-based models such as Vision Transformer (ViT) will significantly improve PCOS detection accuracy and performance compared to traditional methods.
8. The idea that the integration of advanced digital technologies into clinical practice will not compromise patient privacy or data security.

1.5 Limitations

1. The limited availability and quality of scientific data can also affect the accuracy of PCOS diagnosis methods.
2. Changes in PCOS symptoms and presentation may also help improve standardized diagnostic criteria.
3. Interpretive biases in medical imaging and variability among healthcare professionals may have additional implications for the consistency of research.
4. The transformer approach can also be very powerful and requires a lot of data.
5. Accurate diagnosis PCOS screening can have a big impact on patients' health and demographics.
6. Ethical considerations: algorithmic biases, privacy, etc.
7. External influences on PCOS screening.

8. Like with health, long-term impacts on patient outcomes will require further clinical validation studies.

1.6 Delineation

- Define Clear Criteria for PCOS Diagnosis
- Identify Important Trademarks and Criteria Necessary for a Reliable Diagnosis of PCOS, Including Hormonal Imbalance, Uterine Morphology, and Clinical Symptoms
- Establish a Standardized Framework and Framework for PCOS Screening Strategies
- Define Systematic Procedures and Recommended PCOS Diagnosis, Ensuring Accuracy and Reproducibility in a Specific Health System
- Identify and Explain the Limitations and Challenges of Modern Methods
- Identify the Deficiencies and Limitations of Current PCOS Screening Methods, Including Variations in Presentation of Symptoms, Interpretive Bias in Clinical Imaging in 1999, Barriers to Access Information
- Inform the Development of Innovative Solutions to Overcome Limitations
- Utilize the insights gained from this data to drive improvements in New and Effective PCOS Screening Strategy, Including Better Computer Techniques and Assessment Programs
- Improve Screening Accuracy, Coverage, and Health Care Outcomes for Individuals with PCOS
- Finally, Defining PCOS Diagnostic Methods to Improve Screening Accuracy, Optimizing Patient Care Pathways, and Individuals with PCOS for Health outcomes and Targets

1.7 Research questions

Research questions for a study in PCOS detection may include:

1. What test is the maximum appropriate for medical imaging to correctly diagnose PCOS?

2. How can standardized programs enhance uniformity in PCOS prognosis in scientific settings?
3. What is the primary drawback of contemporary PCOS diagnostic techniques and how can they be remedied?
4. How do superior computational fashions, which include transformer-based methods, enhance PCOS detection accuracy as compared to traditional techniques?
5. What moral issues are raised with the aid of advanced diagnostic strategies for PCOS analysis and how can these be addressed?
6. How can transformer-based model integration be optimized for maximum diagnostic effectiveness in medical use?

1.8 Brief chapter overviews

- Chapter 1 Introduction
- Chapter 2 Literature Review and Analysis
- Chapter 3 Methodology
- Chapter 4 Findings and Analysis
- Chapter 5 Conclusion

Chapter 2

Literature Review and Analysis

2.1 Introduction

Polycystic ovary syndrome (PCOS) is the most common health problem affecting women in their reproductive years, 5 to 10% of them have PCOS ^[4]. Here we begin a comprehensive review of the research on PCOS diagnosis. As we move into studies, our goal is to gain a better understanding of the state of PCOS screening methods, including advantages, disadvantages, and potential applications. This review of the literature prepares the ground for us subsequent research by microanalysis and collecting relevant studies.

2.2 Related work in PCOS detection

- CNN-based feature extraction model for PCOS was proposed by Sumathi et al.^[8]. The model is based on image processing feature extraction to detect the cyst or not. The feature extraction watershed algorithm was used, and the calibration parameters were measured using OpenCV. The model can only identify cysts, but cannot classify the presence or absence of PCOS. The accuracy of this model is 85%.
- Kumari ^[9] classified PCOS based on image dataset using DenseNet-121, VGG-19, DenseNet-121, InceptionV3, and Resnet-50. VGG-19 was the most accurate of the four, rising to 70 percent, according to her findings. The author of this study overcame the obstacles of the dataset using the structure of the Generative Adversarial Networks (GANs) and data enhancement. There were a total of 94 images, 50 of which were portraits of PCOS, and the remaining 44 were ultrasound images. None of them depicted PCOS.
- Cahyono et al.^[10] proposed a method where the characteristic extraction is mechanically performed using CNN (100% f1-ratio) and a mean (75.36%) on a 5 overlap cross-validation. The dataset consists of 40 Non-PCO information and 14 Polycystic Ovary Information (PCO).
- Using ResNet 18, U-net and the multinstance learning (MIL), a deep learning architecture was proposed by Lv et al.^[11] to classify PCOS patients using scleral images with an AUC of 98 and an accuracy of 93%. From the 721 images in the dataset, the

proposed deep learning model identified 388 PCOS patients.

- Dewi et al.^[12] proposed a system that incorporates Gabor Wavelet (GWS) strategy for feature extraction and CNN. The first step is to preprocess the images by using various techniques such as: Data cleaning, Image binarization, Morphology, Inverted image, Grayscale, Segmentation, GWS is optimized for feature extraction and lastly, CNN is applied. CNN has the highest accuracy of 80.84% and the time handle is 60.64 seconds.
- Rihana et al.^[13] turned ultrasonic images of ovarian cysts into grey scale images using image binarization as an image preprocessing step. The authors then postprocessed the images by labeling and connecting components, which allowed them to extract geometric features and classify the cysts with 90% accuracy using SVM as a classifier.
- Hossain et al.^[14] developed a CNN-based model called PCONet for PCO detection from ovarian ultrasound pictures. In addition, they improved another CNN model called InceptionV3 (45-layer pre-trained CNN) by using transfer learning to classify ultrasound images of Polycystic ovarian tissue. Using quantitative performance evaluation metrics, they compared PCONet and InceptionV3 and found PCONet outperformed InceptionV3 by 98.12% and 96.56% on test images, respectively.
- Soni and Vashishta ^[15] used image segmentation and CNN (convolutional neural network) to identify PCOS in ultrasound images. K-Nearest Neighbor (KNN) was used to classify ultrasound images.
- According to Bhosale et al.^[16] an LR classifier can be used to detect PCOS. This method is suggested for the automated quality assessment of PCOS data with a DCCN. They proposed a deep learning model that can identify the PCOM of the ultrasound image with 84.81% accuracy when combined with the Inception model.
- Srivastava et al.^[17] used their own dataset to refine the 16 Layered VGG-16 model for PCOS classification from ultrasound images. The model achieved a 92.11% accuracy.
- Alamoudi et al.^[18] proposed a fusion model that combined the ultrasound image and clinical information to determine if the patient had PCOS or not. The best model, combining clinical features and picture feature extraction with MobileNet architecture, achieved an accuracy of 82.46%.

- Purnama et al.^[19] suggested PCOS detection through ultrasound images based on follicle detection. Several classification methods, such as SVM, KNN, etc., were used in this study. The highest accuracy obtained from SVM – RBF kernel was approximately 82.55%.
- Deshpande and Wakankar^[20] proposed an automated method of PCOS detection based on the number of follicles in an ovarian ultrasound picture. Clinical, biochemical, and imaging parameters are then combined to classify patients into two groups: normal and PCOS-affected. Preprocessing, feature selection, segmentation, and classification steps make up the model. The classification process employs the SVM technique.
- A system was proposed by Thara et al.^[21] to classify PCOS by feature extraction using the Structural Normalized Square Similarity detection approach (SNSSDA), and an ANN. Image preprocessing, such as scaling, noise reduction, etc., is first performed. Then, the segmentation approach is applied. The detection of follicles in the ovary picture was done by using adaptive k means clustering techniques. After segmentation, the feature extraction is based on the SNSSDA, and an ANN based classifier is used to identify follicles.
- A new approach to PCOS detection using image processing techniques was introduced in a paper by Maheswari et al.^[22] The main focus of this paper is to eliminate noise in ultrasound images by adaptive histogram equalization and then extract PCOS-related features. Feature Selection is then used to select the features. This approach addresses the shortcomings of traditional methods. The classification is done using Naive bayesian classifier (NBA) and artificial neural network (ANN). The early stage of PCOS detection is 98.63%.
- Using ultrasound imaging, Gopalakrishnan and Iyapparaja^[23] provide a comprehensive approach for PCOS detection. Various image processing techniques are used for preprocessing. Standard methodologies are used to improve image quality. Canny edge detection is used to detect follicle edges. SIFT (Scalable Invariant Feature Transform) algorithm is used for feature extraction to detect the syndrome presence. SVM is used for training and classification. SVM outperforms other benchmarking algorithms with 94.40% accuracy.

- Erdemir and ERDAş^[24] investigate deep learning methods for PCOS detection from ultrasound images to enhance diagnostic efficiency. They examine the performance of XCEPTION, RESNet-152, AND DENSENet-201 model trained and tested with or without data augmentation. They also look at transfer learning. DenseNet-201 achieves the highest accuracy 99.48% with data augmentation. XCEPTION achieves 98.44% with transfer learning, compared to training without transfer learning 75.58%. ResNet-152 initially shows lower performance, but data augmentation improves its accuracy 83.12%. These findings highlight the usefulness of data augmentation and transfer learning techniques in improving PCOS diagnostic accuracy.

2.3 Conclusion

In conclusion, a deep dive into the literature on the detection strategies for PCOS shows that there are a number of exclusive approaches, including deep learning algorithms, CNN-based feature extraction models, image preprocessing strategies with different levels of accuracy, and several approaches have been developed for the diagnosis of PCOS.

Chapter 3

Methodology

3.1 Introduction

In this section, we present the methodology used in our study to diagnose polycystic ovarian syndrome (PCOS) using transformer vision and deep learning techniques. Our study focuses on the need for early diagnosis of PCOS, due to its significant impact on women's health and fertility. We start by outlining the preliminary measures implemented to guarantee the relevance and quality of the data utilized in our investigation. We will next use Vision Transformers (ViT) as our deep learning models for PCOS detection, namely ViT Base and ViT Large. We provide clarification on the experimental design, including the selection of data sets to assess our model's performance, model training techniques, and assessment measures. We want to enhance PCOS with this all-encompassing strategy by utilizing cutting-edge transformer visual deep learning techniques. We can work to make identifying techniques better.

3.2 Research Design

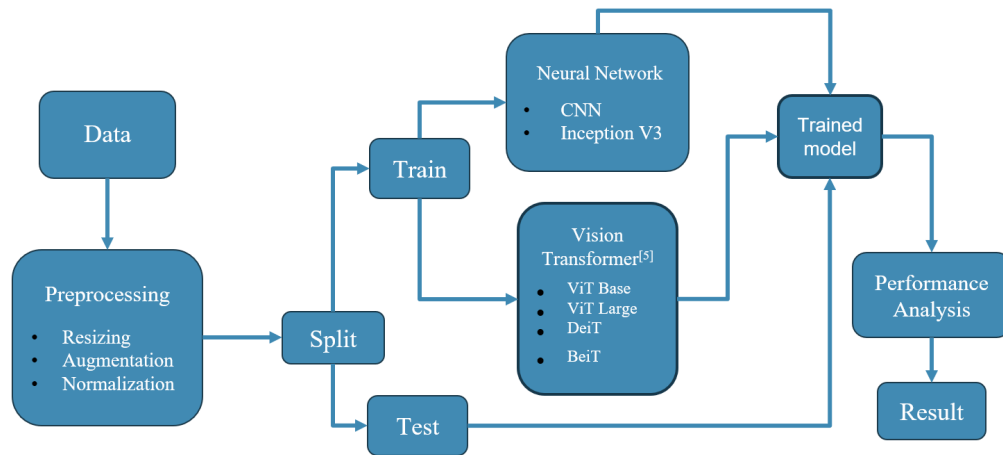


Figure 3.1: Steps in Developing the PCOS Detection Model

3.3 Research instruments

- Python
- Colab
- Hardware implementation

3.3.1 Python

Python is A perceived, object-situated programming language very much like PERL, that has acquired quality because of its reasonable linguistic structure and lucidness. Python is professed to be nearly clear to find out and moveable, and that implies its assertions are perceived in a very assortment of usable frameworks, as well as UNIX-based frameworks, Mac OS, MS-DOS, OS/2, and various variants of Microsoft Windows ninety-eight. The creator of Python was Guido van Rossum, a previous occupant of the Kingdom of The Netherlands, whose most loved parody bunch at the time was Monty Python's Flying Circus. The ASCII text record is uninhibitedly available and open for adjustment and use. Python incorporates a significant assortment of clients. The accompanying libraries of python were utilized all through this exploration.

- **NumPy:** NumPy, which represents Numerical Python, is a library comprising multi-layered cluster objects and an assortment of schedules for handling those exhibits. NumPy can be used to generate numerical and intelligent methods in the displays. NumPy is a package for Python, created by Travis Oliphant in 2005. It is an open-source work that anyone can use without any restrictions.
- **Pandas:** Pandas is an open-source Python package widely used for AI, informatics, and information analytics projects. It is based on top of another bundle named NumPy, which offers help for multi-layered clusters.
- **Scikit-learn:** Scikit-learn is an open-source Machine Learning Python bundle that offers usefulness supporting supervised and unsupervised learning. Also, it gives tools to show improvement, determination, and assessment as well as numerous different utilities including information pre-processing usefulness.

- **Resample:** The `resample()` function is utilized to resample time-series information. It is an accommodation technique for recurrence change and resampling of time series. The object should have a datetime-like record (`DatetimeIndex`, `PeriodIndex`, or `TimedeltaIndex`), or pass Date-Time-like qualities to the `on` or `level` keyword.
- **SpaCy:** SpaCy is a Python natural language processing open-source library. It is designed to create general language measurement lists or data extraction lists and is built in Python. It functions and offers an intuitive API along with a brief for everyday use.
- **ScispaCy:** ScispaCy is a Python bundle consisting of spaCy models for handling biomedical, logical, or clinical text.
- **TensorFlow:** TensorFlow is a Python library for quick mathematical processing made and released by Google. Deep Learning models can be created with an establishment library directly or by using covering libraries that improve on the cycle based on top of TensorFlow.

3.3.2 Colab

Collaboratory, sometimes referred to as "Colab", is a Google Research product. Collaboration It is particularly useful for machine learning, data analysis, and education because it is. It allows anyone to write and run arbitrary Python code through the browser. Technically, Colab is a hosted Jupiter notebook service that offers it for free. No programming is required to access and use computing resources, including the GPU.

3.3.3 Hardware implementation

The following hardware tools (Colab) were utilized for implementation:

1. Colab CPU
2. 12 GB Virtual RAM
3. 12GB NVIDIA Tesla K80 GPU

Generally, RAM is used to store frequently accessed and easily accessible data or files. Such RAM configurations are available in 4, 6, and 8 GB variants for a normal CPU. when The

model was trained on a small data set that fit easily in system RAM. However, if the system RAM is not large enough to train the model with such a large data set, the data set can be very large. Google Collab has provided 12 GB of free virtual RAM for free to fix the problem.

3.3.4 Datasets

In our research on Polycystic Ovary Syndrome (PCOS) detection using Transformer-Vision and Deep Learning methods, we use mainly ultrasound image data. Ultrasound images: This dataset collects high-resolution ultrasound images of uterine tissue from patients diagnosed with PCOS. Each image shows the internal structure of the uterus and can reveal symptoms related to PCOS, such as cysts, follicles.

Dataset Vsualization

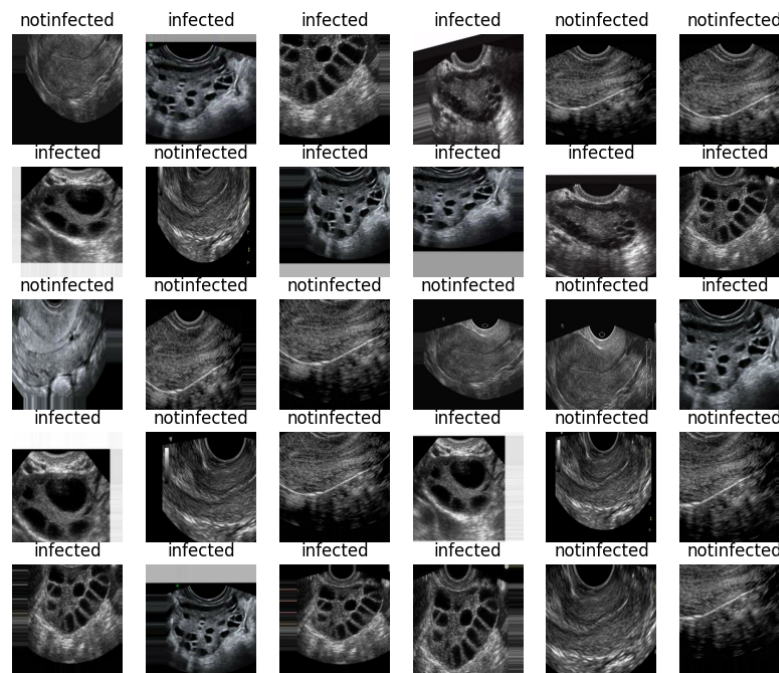


Figure 3.2: Dataset Vsualization

3.3.5 Data pre-processing

For PCOS detection, we preprocess ultrasound images in various ways before incorporating them into our Transformer-Vision and Deep Learning models. This assures the accuracy and quality of the data. These actions are:

1. Image size: In order to normalize the image data for our models, we have reduced all the images of both the data sets to 224×224 px. Each of the training, test and validation sets consists of 16 batches.
2. Data Enhancement: To increase the robustness and generalizability of our model, we use data enhancement techniques like rotation, flipping, zoom, etc., which are used to artificially increase the diversity of training data. This helps prevent overfitting and enhances the model's ability to generalize to unseen data.
3. Normalization: To improve the consistency and convergence of the training during optimization, we normalize the pixel values in the ultrasound image to a standardized range (e.g., [0, 1] or [-1, 1]) using Keras ImageDataGenerator.

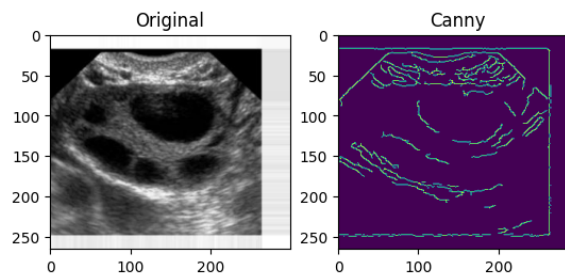


Figure 3.3: Edge Detection

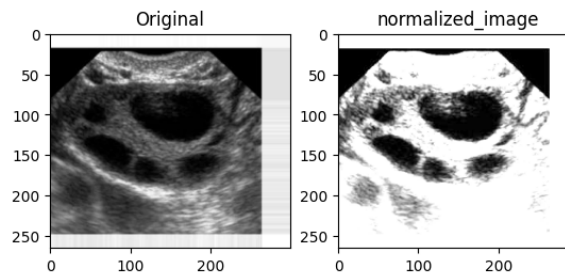


Figure 3.4: Binarization

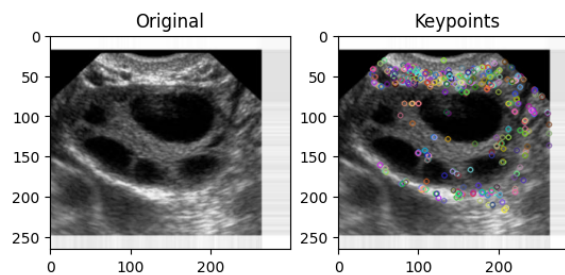


Figure 3.5: ORB

3.3.6 Feature extraction

In our research on PCOS detection using transformer vision and deep learning methods, feature extraction refers to the process of capturing meaningful ultrasound images necessary to differentiate between PCOS positive and PCOS negative cases.

1. Image embeddings: We extract image embeddings from ultrasound images preprocessed using ViT models, which capture semantic information in images and are vector representations with higher dimensions. These embeddings encode features such as textures, shapes, and spatial relationships, discrete patterns associated with PCOS, and the ability to learn.
2. Transformer layers: ViT models stack transformer layers, which process image embeddings through focused methods to capture global and local dependencies in the image. Each transformer layer attends to appropriate image areas, gathering information from neighboring pixels and fixing the position.
3. Fine-tuning: We adjust the parameters of pre-trained ViT models in our dataset to maximize their performance in PCOS identification. In the process of fine-tuning, the model applies feature exclusion and label information about snowflakes to learn how to better tune its internal region to differentiate between ultrasound images with and without PCOS.

3.3.7 Splitting into training and testing

Train	Test
2400	244

Table 3.1: Splitting into training and testing

3.4 Random over-sampling

Random oversampling is a technique used to deal with class imbalance in a dataset, especially when one class (e.g., PCOS-positive cases) is underrepresented relative to another (e.g., PCOS-negative cases) Ultrasound imaging in our analysis of PCOS detection -Class imbalance can be encountered in the data set, where the number of PCOS-positive reflections is much less than the number of PCOS-negative reflections

To reduce this imbalance and ensure that our deep learning models are trained on balanced data, we use random oversampling. This method allows a random sample to be selected from a subclass (PCOS-positive) and repeated twice to increase the dataset until the class distribution is balanced

The process of random oversampling involves the following steps.

1. Identify subgroups: Identifies underrepresented groups in the dataset, in this case the group including PCOS.
2. Randomly Select Samples: Randomly select samples in the alternative PCOS-positive category. This means that some samples can be selected repeatedly for inclusion in the augmented dataset.
3. Second example: A selected sample with PCOS was included in the dataset, effectively increasing the number of people with PCOS.
4. Repeat until equilibrium: Continue the process of random selection and duplication until the class distribution is equilibrated, with the same number of samples in each class.

3.5 Analysis

Various deep learning algorithms have been used to diagnose polycystic ovary syndrome (PCOS), each offering unique advantages and capabilities. The major deep learning algorithms used in PCOS diagnosis include:

3.5.1 CNN

Convolutional Neural Networks (CNNs)^[32]: The CNN is based on the brain's visual cortex, which is comparable to the eye in humans. CNN extracts knowledge from the input image by using a subset of the image. In a fully connected network, the learning phase uses multiple parameters. For instance, RGB color images with a width of 300 pixels and a height of 300 pixels would create a network with 270,000 at the learning phase. These networks are massive in size. In contrast, convolution operation allows CNNs to significantly reduce the number of parameters by applying a kernel or a filter window to the input images.^[25]

There are three elements to CNN

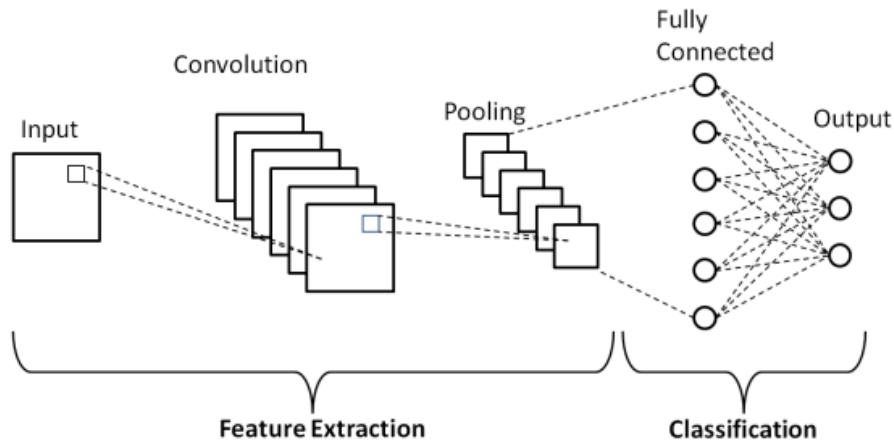


Figure 3.6: A basic convolutional neural network (CNN) architecture. [26]

1. **Input layer:** The input data is represented as a four-dimensional tensor on the input layer. It represents: the number of images, the width of the image, the height of the image, and the number of channels, where the number of channels is one for a grayscale image and three for a colored image.
2. **Learning layer or convolutional layer:** The learning layer is the core of CNN. It is a point product between the regions of neurons of the input layer. The weight matrix of the input layer is a point product. A filter or kernel is then created in the network to output the value for the region. Typically, several convolution layers are used to detect features. It teaches positions.
3. **Classification layer:** The classification layer is a fully connected layer with a neuron for each class. The output is a probability that this input belongs in this class.

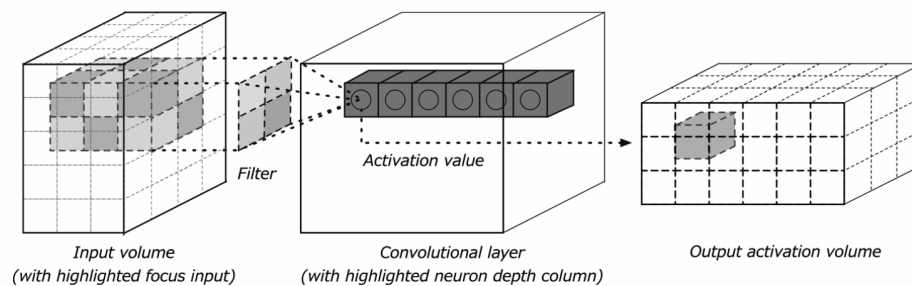


Figure 3.7: Convolution layer. [25]

CNN may also have other layers in its learning layer and output layer, such as:

Pooling layer

Pooling layers come after convolution layers in a neural network. Their job is to shrink the size of the data being processed, which helps prevent the network from memorizing too much and becoming overfit. Pooling also reduces the amount of data by taking only the most important information. Generally, pooling takes place in 2x2 pixels blocks that move by 2 pixels every time. There are two types of pooling. Max pooling and average pooling. Max pooling keeps the highest value in every block. Average pooling keeps the average value of every block.

Dense layer or fully connected layer

In addition to convolutional and other layers, dense layers are used at the end of the learning layers. and so on It is used to calculate the class score at the output layer where the output will be $[1 \times 1 \times N]$; N is the number of classes. At this level the input volume and common parameters (weights and biases) are to compute each neuron.

3.5.2 Inception V3

Inception V3 is a widely used Convolution Neural Network (CNN) structure that is commonly used for image classification. Invented in 2015, the inception v3 model has 42 layers and a low error rate compared to previous generations.^[27] Inception V3 is also referred to as a Convolution, Pooling, Deep Convolution Neural Network with multiple layers of Fully Connected Layers. The idea of “start modules”, which are parallel convolution layers with varying filter sizes, allows the model to capture more local and global information from the input images.

Inception V3 can also be used for more than just image classification, such as object search, classification, and natural language processing tasks. Its flexibility and power are unmatched, making it one of the most important tools for deep learning.

3.5.3 The Transformer: Attention is all you need

In 2017, the Google Brain team released a paper called “**Attention is all you need**”^[29] which described and introduced a novel architecture called **Transformer** for language modeling. Another tool, *attention*, was explored for filter extraction. This is a tricky one Specifically, it relies on a dynamic filter that can be slightly weighted Specific features of the input sequence relevant to the prediction task. The main thing The advantage of such a construction is that

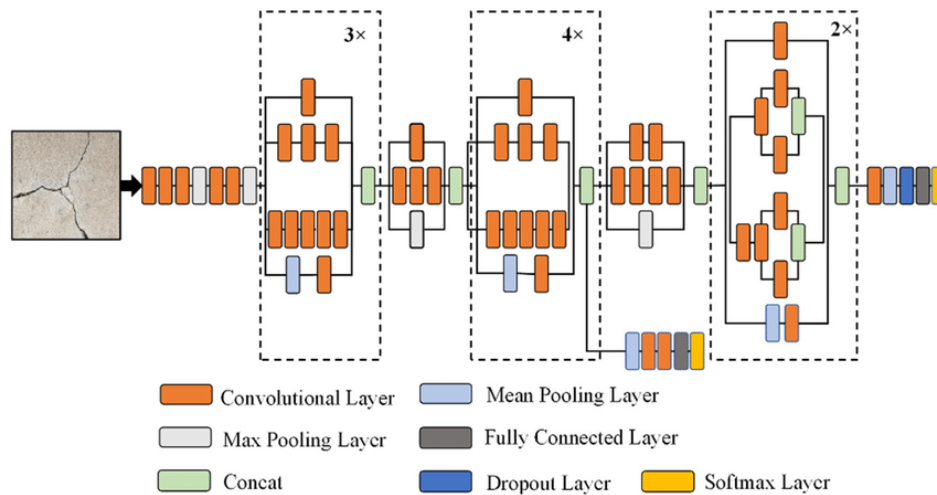


Figure 3.8: Inception V3 architecture.^[28]

it can be mathematically equated As for the gold standard architecture of the time, Recurrent Neural Networks (which sequentially analyzed sentences and took much longer to train).

The transformer is made up of two main parts:

- **Encoder:** Processes the input words in a batch (in parallel).
 - **Word encoding:** Search a dictionary and convert each word into a token, i.e., an integer value. At the end, the entire input text is represented as an array.
 - **Word embedding:** We use a linear transformation to project each token in the d-dimensional space that represents the primary semantic properties of each word in our input array. For example, the vector that encodes “actor” will be very similar to the vector encoding “actress,” with a difference in the dimension representing gender.
 - **Positional embedding:** In addition to a word’s meaning, we also want to know where it is in the text. Otherwise, sentences containing the same words but different word order could be confusing. To indicate each word’s position in the text, we append additional numbers to its set. This improves our comprehension of sentences.
 - **Multi-headed self-attention layer:** After the embedding stage, each Multi-Head self attention layer consists of three steps: The first step is the core data (**values**). The second step is the object to compare with (**keys**). The third step is the **query**. Let’s look at an example: We’re searching YouTube for a video. We enter

keywords in the search field. This is our question: Next, we use the video titles as keys. YouTube compares our query with the video titles. If the video title matches our query, we view the video's content (the value). Once we have calculated these vectors, we can use them to calculate the attention using the formula below:

$$\text{Attention}(Q,K,V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

where Q , K , and V are respectively the matrices of the computed queries, keys, and values, and d_k is the dimension of the key vectors, which is used as a normalization factor. The attention filter, computed with the dot product QK^T , is a matrix that encodes the importance of each word with respect to all the other words in the text (hence, the name **self-attention**), and its product with the value matrix computes a filtered version of the sentence.

A single attention filter captures the information about a single idea. To understand the different ideas of the input we need to compute

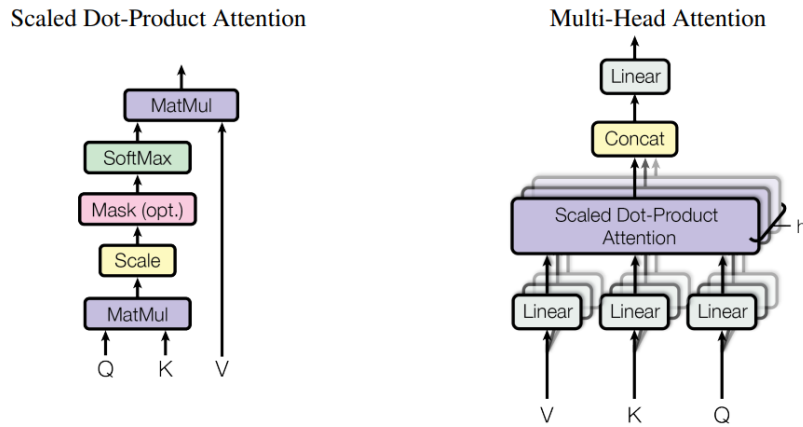


Figure 3.9: Single and multi-headed self-attention architecture.^[29]

attention with multiple independent attention filters. This part of the process is called multi-head self-attention. Following the multidimensional self attention computation, each head's filtered version of the input is enriched with the original data using concatenation, skip connections, and normalization. This version of the input is then passed to the fully connected self attention layer, which forms the tokens with the necessary dimension through a linear transformation before sending the processed input to the subsequent multidimensional self attention layer.

- **Decoder:** Once the data has been encoded, the encoder interprets it into a series

of tokens containing semantic and context-sensitive information. The decoder then analyzes these tokens to forecast the outcome, which may include a list of the words in the translated text. It then compares the predictions with the ground truth labels to estimate the loss.

- **Masked Multi-Head Self-Attention:** In the decoder's first step, each target position (the words in the right output sequence) responds to all the positions in the input sequence, but it's important to avoid "cheating" by allowing a position to respond to future input positions while training. To overcome this, we use a masking mechanism. This means that each position in the target sequence can only respond to the positions that precede it in the input sequence and the model is auto-regressive, i.e., it doesn't peek into the future.
- **Add and Norm:** Following self-attention layer normalization is implemented and residual connections are added. This phase aids in training stabilization.
- **Encoder-Decoder Cross-Attention with Masking:** The model must build connections with the source sequence of the encoder's output while complying by the causal mask in order to preserve the autoregressive behavior of the decoder in the second stage. The "multi-head cross-attention" cross-attention mechanism uniquely does this. Similar to self-attention, cross-attention begins with masking. But here, disguising has a different purpose. Rather of stopping future places in the target sequence from being attended, masking keeps the attention from being directed toward the padded or out-of-the-way spots in the goal sequence. In order to preserve the auto-regeneration feature of the model while concentrating on the pertinent data in the source sequence, masking is used when building the target sequence.
- **Add and Norm:** The feed-forward layer's output is processed using residual connections and layer normalization.
- **Feed Forward:** Position wise feed-forward networks perform a linear transformation and then a non- linear activation function on each position individually.
- **Add and Norm:** Similar to the preceding step, residual connections and layer normalizations are applied to the feed-forward output.
- **Linear Layer:** This layer is responsible for connecting the model's internal

representation to the output dimension.

- **Softmax:** Probability distribution of output vocabulary is given by softmax function. The model can predict the following token in series. To sum up, transformer decoder is designed for sequence output generation with self-attention layer, feed-forward layers, causal generation and sequence dependencies.

3.5.4 Vision Transformers

In 2021, in a paper titled “**An image is worth 16x16 words: Transformers for image recognition at scale**”, Alexey dosovitsky and his team^[30] proposed to use the architecture of the Transformers for image analysis away from its traditional use in text processing.

The basic idea behind this study is that by dividing an image into square patches and embedding them in a vector space, similar to the approach outlined in the preceding subsection, they can be fed into a transformer encoder that builds intermediate representations. These representations can then be analyzed by a classifier (e.g., a multi-layered perceptron) to predict the label.

The proposed architecture is called the Vision Transformer, or ViT, and it uses an encoder transformer to classify images.

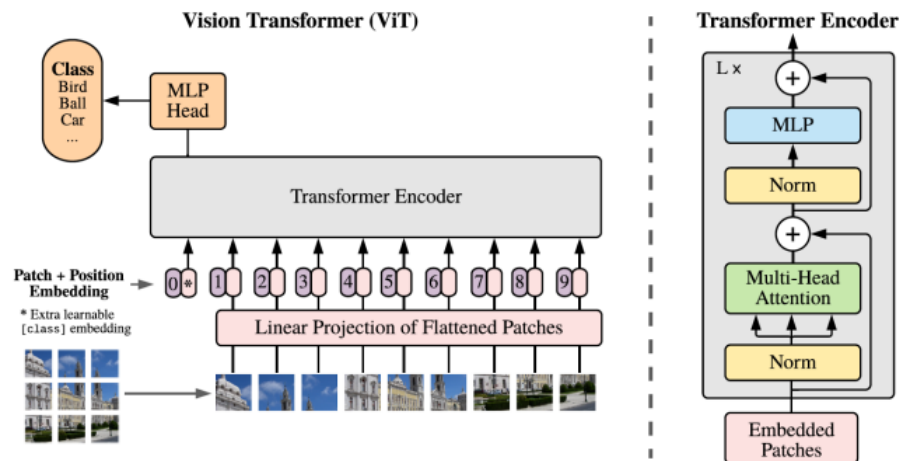
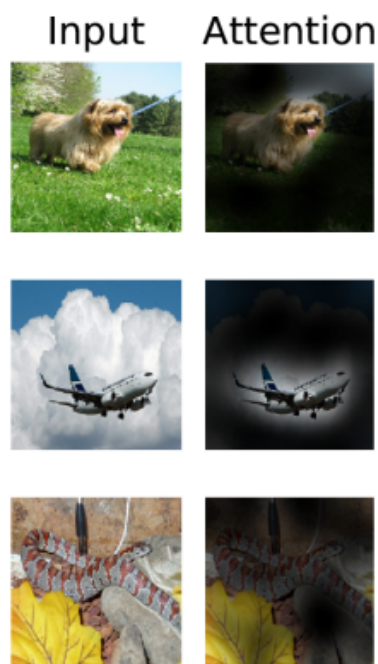


Figure 3.10: Architecture of the Vision Transformer. ^[30]

Using a Vision Transformer (ViT) instead of a Convolutional Neural Network (CNN) has numerous benefits, such as:

- **Handling Long-Range Relationships:** Vision Transformers specialize at handling long-distance relationships in pictures because they feature a self-attentional mechanism. This indicates that they are able to capture global correlations between distant patches of pixels. CNNs frequently face difficulties with this.^[31]

In addition, self-attention allows Vision Transformers to accurately identify the dependencies of every pixel in the image, no matter how close they are to each other in space. Because of their comprehensive approach, Vision Transformers are ideal for applications that require a deep understanding of visual context, allowing them to capture complex relationships and patterns across a picture.



In a CNN, the input image is examined by swiping a cluster of kernels over the whole image in a localized area. This implies that the convolution in CNN design is only able to get global information in the top layers, while the convolution is done across adjacent pixels in the bottom layers. No matter how close each pixel or patch is to each other, each pixel or patch in the ViT can attend to each other simultaneously because it is self-attended. This is an important advantage for computer vision jobs because it enables ViTs to understand the relationship between different parts of an image. For example, a ViT can capture the interactions between distant horizon, sky, coastline, and people in a beach scene.

Figure 3.11: Attention maps.^[30]

This holistic view allows ViTs to identify patterns, contexts, and relationships that may otherwise be overlooked by conventional CNNs that rely on local element extraction through convolution filters.

- **Scalability:** Self-attention is at the heart of ViTs. One of its main features is that ViTs can handle a large number of picture sizes. In real life, images come in many different sizes. Each input element, like a token or a pixel, talks to all the other elements in the ViT network at the same time. Because of this unique function, we can specify

the exact proportions of an input image in advance. But by dynamically changing the input's dimensions, the network provides unparalleled flexibility. This flexibility is helpful when there are variations in image size or resolution.

- **Greater Parallelism:** ViTs can perform parallel attention across all image positions, resulting in higher computational efficiency, particularly on hardware accelerators such as GPUs or TPUs, resulting in faster training and inference time. This allows ViTs to perform large-scale and complex tasks with high efficiency.

In the initial paper ^[30], it has been demonstrated that the architecture generally outperforms the CNN-based architecture of ResNet (with lower computational budget) pre-trained on large datasets (see Appendix A for more information on transfer learning). In Figure 3.8, that compares several large ViT models that have been pre-trained on JFT-300M dataset ((that comprises more than 300 millions of images, and which is not publicly available) and ImageNet21k dataset (which contains 14 million images) to a larger ResNet model. We can see that although the number of pre-training TPUv3 cores spent on ViTs is significantly smaller compared to the architecture of ResNet, the latter is greatly outperformed by the Vision Transformers pre-trained on a JFT dataset.

	Ours-JFT (ViT-H/14)	Ours-JFT (ViT-L/16)	Ours-I21k (ViT-L/16)	BiT-L (ResNet152x4)	Noisy Student (EfficientNet-L2)
ImageNet	88.55 ± 0.04	87.76 ± 0.03	85.30 ± 0.02	87.54 ± 0.02	88.4/88.5*
ImageNet ReaL	90.72 ± 0.05	90.54 ± 0.03	88.62 ± 0.05	90.54	90.55
CIFAR-10	99.50 ± 0.06	99.42 ± 0.03	99.15 ± 0.03	99.37 ± 0.06	—
CIFAR-100	94.55 ± 0.04	93.90 ± 0.05	93.25 ± 0.05	93.51 ± 0.08	—
Oxford-IIIT Pets	97.56 ± 0.03	97.32 ± 0.11	94.67 ± 0.15	96.62 ± 0.23	—
Oxford Flowers-102	99.68 ± 0.02	99.74 ± 0.00	99.61 ± 0.02	99.63 ± 0.03	—
VTAB (19 tasks)	77.63 ± 0.23	76.28 ± 0.46	72.72 ± 0.21	76.29 ± 1.70	—
TPUv3-core-days	2.5k	0.68k	0.23k	9.9k	12.3k

Figure 3.12: Transfer of large datasets on large models.^[30]

When ViT is trained on the smaller dataset size of ImageNet21k, the ViT model produces similar results to the CNN architecture, but with only about 1/40th the computational power. On the other hand, the following figure shows that ViTs are unable to compare with ResNet models when pre-trained on a smaller dataset size of 1K classes (1.3 million images). In addition, the image shows that with a larger dataset size, the size becomes a major factor in the performance of the model, and larger models are more accurate.

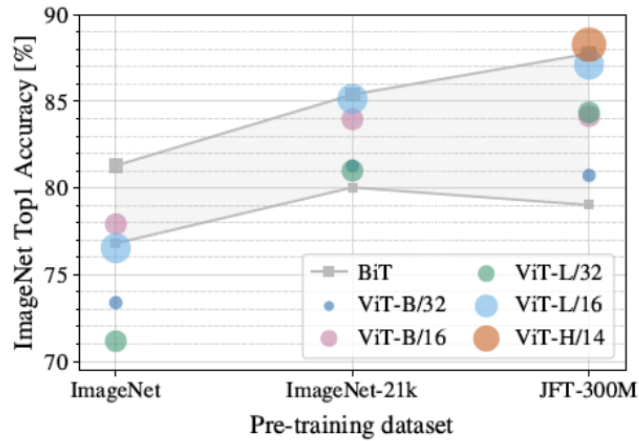


Figure 3.13: ViT vs CNN Pre-training Dataset Size Comparison.^[30]

ViT base

ViT models have gained a lot of attention for their performance in image classification. ViT base, trained with a 16x16 pixel patch size and 224x224 pixel input image size, is an example of an optimized ViT architecture. Unlike CNNs, ViT models take a self-directed approach to capturing global and local context in image input. By doing so, ViT models are able to process spatial and temporal information more effectively, making them ideal for applications that need a comprehensive understanding of image content.

ViT Large

The ViT Large represents the part of the ViT framework that is optimized for image processing. The ViT Large is significantly larger than the ViT base model. It has more transformer levels and more parameters. The enhanced sampling capability of ViT Large allows for more complex shapes and signals to be captured from embedded images. This makes ViT Large a great choice for projects that demand a deep understanding of image content.

DeiT

Data-efficient image Transformers (DeiT)^[33] are transformer-based models trained only on ImageNet and trained in less than 3 days on just one computer. Despite its compact size (86 million parameters), DeiT has achieved top-1 accuracies on ImageNet with no external data. DeiT uses a unique teacher-student training strategy specifically designed for transformers and includes a distillation variant that treats a teacher's hard decision as a real label. One of

the unique features of DeiT is its new distillation token, which interacts with class tokens and patch tokens via self attention layers.

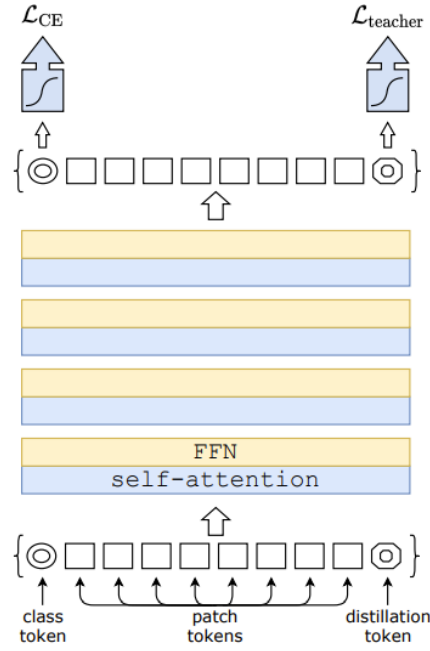


Figure 3.14: DeiT Architecture.^[33]

BEiT

BEiT^[34], short for BERT Pre-Training of Image Transformers, utilizes the bidirectional encoder representation of Image Transformers. The BEiT algorithm trains vision Transformers through a masked image modeling task. Each image is examined in two perspectives: as an image patch (16x16 pixels). During the pre-training phase, the model reconstructs the original visual token using the image patch. For classification tasks, additional layers are appended to the pretrained encoder. By learning strong representations from images, BEiT demonstrates improved performance across a range of downstream tasks.

3.5.5 Hyperparameter tuning

A hyperparameter is a parameter that is set before training begins. It is not a parameter that is learned during training. A hyperparameter controls the behavior of a learning algorithm. It can have a big effect on a deep learning model.

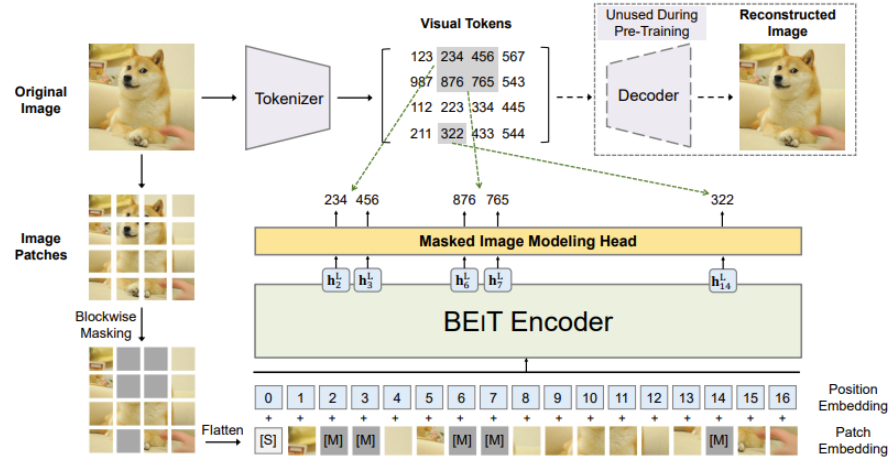


Figure 3.15: BEiT Architecture. [34]

3.5.6 Limitations

When deep learning models are used to diagnose PCOS (polycystic ovary syndrome) from ultrasound images, there are several limitations to consider:

1. **Data availability:** The main challenge is to obtain an ultrasound image dataset with written specifications for training and analysis. Large datasets with a large number of PCOS diagnostic imaging scenarios are needed for general integration of strong deep learning characteristics. These kinds of data may, however, be difficult to obtain due to personnel limitations, confidentiality issues, and delays in participation due to late submissions.
2. **Data quality:** The ultrasound image's quality and performance can be affected by noise levels, resolution variations, and other factors.

3.5.7 Ethics

Ethical considerations play an important role in developing deep learning models to detect polycystic ovary syndrome (PCOS) from ultrasound images. Several ethical issues need to be addressed to ensure these models play a responsible and ethical role in clinical practice:

1. **Patient privacy and confidentiality:** Working with medical image data requires researchers to comply with data protection laws and to obtain informed consent for the use of patients' data in research. It is important to keep patient information anonymous and to implement strong data protection measures to protect patients' privacy rights.

2. **Bias and Non-Bias:** Deep learning models can inadvertently create bias in training data which can lead to biased or discriminatory outcomes, particularly in underrepresented populations or less representative patient populations. To ensure an accurate and unbiased diagnosis of PCOS across diverse patient populations, researchers need to minimize biases in data as well as model predictions.
3. **Transparency and accountability:** To build trust between health care providers and patients, it is important to provide explicit documentation on model development, analysis and implementation, as well as training methods and validation results to understand and analyze models. Researchers should calculate and evaluate options that address ethical concerns or unintentional consequences of sampling.
4. **Clinical Validation and Validation:** Prior to clinical design, validation and rigorous assessment against established clinical stains must be performed in a robust diagnostic model.

3.6 Conclusion

In conclusion, deep learning models in medical imaging are a very comprehensive way to diagnose PCOS (polycystic ovarian syndrome) from ultrasound images; However, in this particular model, it is important to understand its limitations and ethical concerns.

Chapter 4

Finding and Analysis

This chapter focuses on accuracy, precision, . Provides recall, f1-scores and training-validation curve and comparison Analysis between state-of-the-art models

4.1 Precision, Recall and F1-scores

1. Precision: A positive predictive value is called the fraction of the precision. The relevant pattern in the retrieved pattern is the symbol of the device learning model's performance – good prediction quality. For example.
2. Remainder: A fraction of the total number is called a sensitivity equation. Relevant samples actually taken reflect actual numbers. True positive reviews (earned) and how many perfect hits.
3. F1 Prizes: F1 prizes are based on accuracy and remember. It is used as a statistical measure of performance.

4.1.1 CNN Classification Report

	precision	recall	f1-score	support
Class 0	0.94	0.94	0.94	122
Class 1	0.94	0.96	0.95	122
accuracy			0.95	244
macro avg	0.95	0.95	0.95	244
weighted avg	0.95	0.95	0.95	244

Figure 4.1: CNN Classification Report

In the classification report, the CNN model displays its performance metrics across both classes. With high accuracy, recall, and an F1-score, the CNN model is very reliable in classifying both class infected and class not infected instances. With an overall accuracy of 95%, the CNN model is highly reliable in classifying instances from the dataset.

4.1.2 Inception V3 Classification Report

	precision	recall	f1-score	support
Class 0	0.9656	0.9590	0.9623	122
Class 1	0.9623	0.9697	0.9660	122
accuracy			0.9645	244
macro avg	0.9640	0.9644	0.9642	244
weighted avg	0.9640	0.9645	0.9642	244

Figure 4.2: Inception V3 Classification Report

In the classification report of Inception V3, the model's performance metrics across both classes are highlighted. With precision, recall, and F1-score values close to or above 0.96 for both Class 0 and Class 1, Inception V3 demonstrates a high level of accuracy in classifying instances from the dataset. The overall accuracy of 96.45% further confirms the model's reliability in accurately classifying instances across the dataset.

4.1.3 ViT-Base Classification Report

	precision	recall	f1-score	support
Class 0	0.9804	1.0000	0.9901	122
Class 1	1.0000	0.9836	0.9918	122
accuracy			0.9868	244
macro avg	0.9902	0.9918	0.9910	244
weighted avg	0.9910	0.9918	0.9910	244

Figure 4.3: ViT-Base Classification Report

In the classification report for ViT Base, the model's exceptional performance metrics across both classes are highlighted. With precision, recall, and F1-score values exceeding 0.98 for both classes, ViT Base demonstrates a high level of accuracy in classifying instances from the dataset.

both Class 0 and Class 1, ViT Base demonstrates outstanding accuracy in classifying instances from the dataset. The overall accuracy of 98.68% underscores the model's robustness in accurately classifying instances across the dataset, making it highly reliable for classification tasks.

4.1.4 DeiT Classification Report

	precision	recall	f1-score	support
not_infected	0.98	1.00	0.99	122
infected	1.00	0.98	0.99	122
accuracy			0.99	244
macro avg	0.99	0.99	0.99	244
weighted avg	0.99	0.99	0.99	244

Figure 4.4: DeiT Classification Report

In the classification report for DeiT, the model's exceptional performance metrics across both classes are highlighted. With precision, recall, and F1-score values at 0.99 for both Class 0 and Class 1, DeiT demonstrates outstanding accuracy in classifying instances from the dataset. The overall accuracy of 99% underscores the model's robustness in accurately classifying instances across the dataset, making it highly reliable for classification tasks.

4.1.5 BeiT Classification Report

	precision	recall	f1-score	support
not_infected	0.98	1.00	0.99	122
infected	1.00	0.98	0.99	122
accuracy			0.99	244
macro avg	0.99	0.99	0.99	244
weighted avg	0.99	0.99	0.99	244

Figure 4.5: BeiT Classification Report

In the classification report for BeiT, the model's exceptional performance metrics across both classes are highlighted. With precision, recall, and F1-score values at 0.99 for both Class 0 and Class 1, DeiT demonstrates outstanding accuracy in classifying instances from

the dataset. The overall accuracy of 99% underscores the model's robustness in accurately classifying instances across the dataset, making it highly reliable for classification tasks.

4.1.6 ViT-Large Classification Report

	precision	recall	f1-score	support
Class 0	0.9804	1.0000	0.9901	122
Class 1	1.0000	0.9836	0.9918	122
accuracy			0.9918	244
macro avg	0.9902	0.9918	0.9910	244
weighted avg	0.9910	0.9918	0.9910	244

Figure 4.6: ViT-Large Classification Report

In the classification report for ViT Large, the model's exceptional performance metrics across both classes are highlighted. With precision, recall, and F1-score values exceeding 0.98 for both Class 0 and Class 1, ViT Large demonstrates outstanding accuracy in classifying instances from the dataset. The overall accuracy of 99.18% underscores the model's robustness in accurately classifying instances across the dataset, making it highly reliable for classification tasks.

4.1.7 Conclusion

In conclusion, the analysis of Vision Transformer (ViT) models including ViT Base, DeiT, BeiT and ViT Large using traditional CNN-based methods such as InceptionV3 provided encouraging results for the diagnosis of polycystic ovary syndrome (PCOS) . Although all performance measures such as accuracy and F1 scores showed satisfactory results. All things considered. The results show how deep learning models can efficiently detect PCOS-related symptoms in ultrasound images, providing important information to enhance diagnostic match.

4.2 Confusion Matrix

The confusion matrix is a table of a distribution systems. Compares the predicted squares with the actual squares for each example In the testing process.

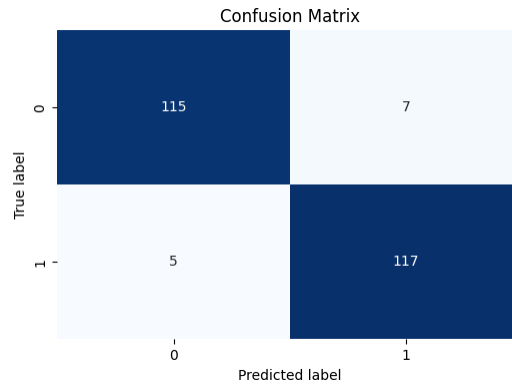


Figure 4.7: CNN Confusion Matrix

The confusion matrix of the CNN model reveals its performance in classifying the dataset. The model correctly classified 115 instances of Class 0 and 117 instances of Class 1. There were 7 instances of Class 0 misclassified as Class 1, and 5 instances of Class 1 misclassified as Class 0. This matrix showcases the model's precision in distinguishing between the two classes, indicating its robust classification performance.

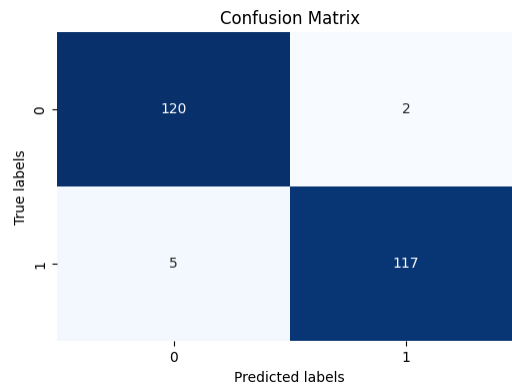


Figure 4.8: Inception v3 Confusion Matrix

The confusion matrix of the Inception V3 model reveals its performance in classifying the dataset. The model correctly classified 120 instances of Class 0 and 117 instances of Class 1. There were 2 instances of Class 0 misclassified as Class 1, and 5 instances of Class 1 misclassified as Class 0. This matrix showcases the model's precision in distinguishing between the two classes, indicating its robust classification performance.

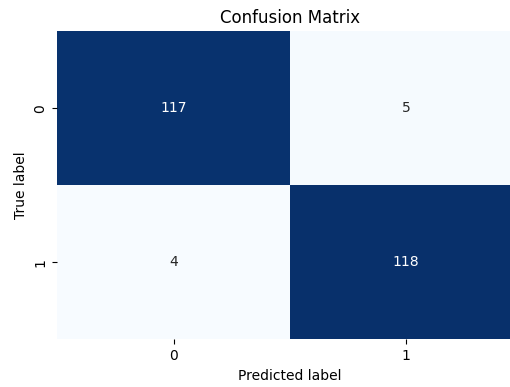


Figure 4.9: ViT Base Confusion Matrix

The confusion matrix of the ViT Base model reveals its performance in classifying the dataset. The model correctly classified 117 instances of Class 0 and 118 instances of Class 1. There were 5 instances of Class 0 misclassified as Class 1, and 4 instances of Class 1 misclassified as Class 0. This matrix showcases the model's precision in distinguishing between the two classes, indicating its robust classification performance.

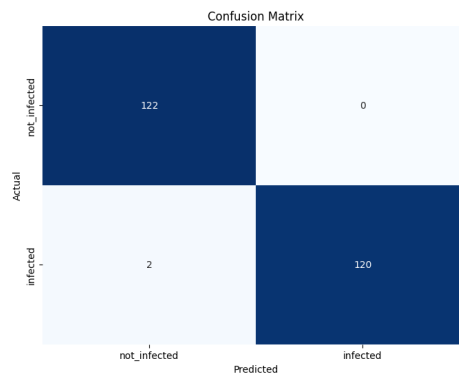


Figure 4.10: DeiT Confusion Matrix

The confusion matrix of the DeiT model reveals its performance in classifying the dataset. The model correctly classified 122 instances of Class 0 and 120 instances of Class 1. There were 0 instances of Class 0 misclassified as Class 1, and 2 instances of Class 1 misclassified as Class 0. This matrix showcases the model's precision in distinguishing between the two classes, indicating its robust classification performance.

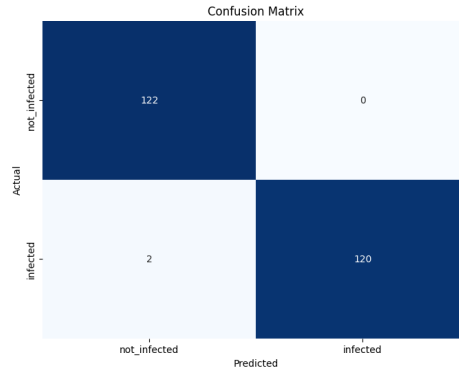


Figure 4.11: BeiT Confusion Matrix

The confusion matrix of the DeiT model reveals its performance in classifying the dataset. The model correctly classified 122 instances of Class 0 and 120 instances of Class 1. There were 0 instances of Class 0 misclassified as Class 1, and 2 instances of Class 1 misclassified as Class 0. This matrix showcases the model's precision in distinguishing between the two classes, indicating its robust classification performance.

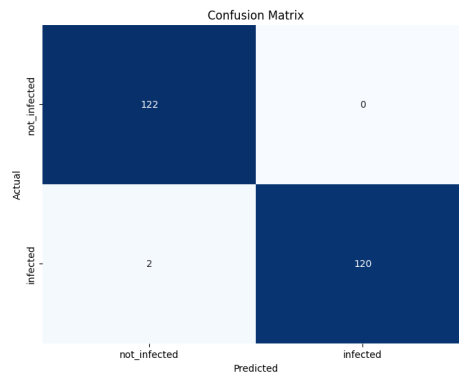


Figure 4.12: ViT Large Confusion Matrix

The confusion matrix of the ViT Large model reveals its performance in classifying the dataset. The model correctly classified 122 instances of Class 0 and 120 instances of Class 1. There were 0 instances of Class 0 misclassified as Class 1, and 2 instances of Class 1 misclassified as Class 0.

4.2.1 Conclusion

In conclusion, the confusion matrix for PCOS detection task provides a detailed breakdown of the performance of the model.

4.3 Train-Validation loss curves

- **Training Loss:** The training loss metric assesses how well a deep learning model fits the training data.
- **Validation Loss:** Conversely, validation loss is a metric used to gauge the performance of a deep learning model on the validation set.

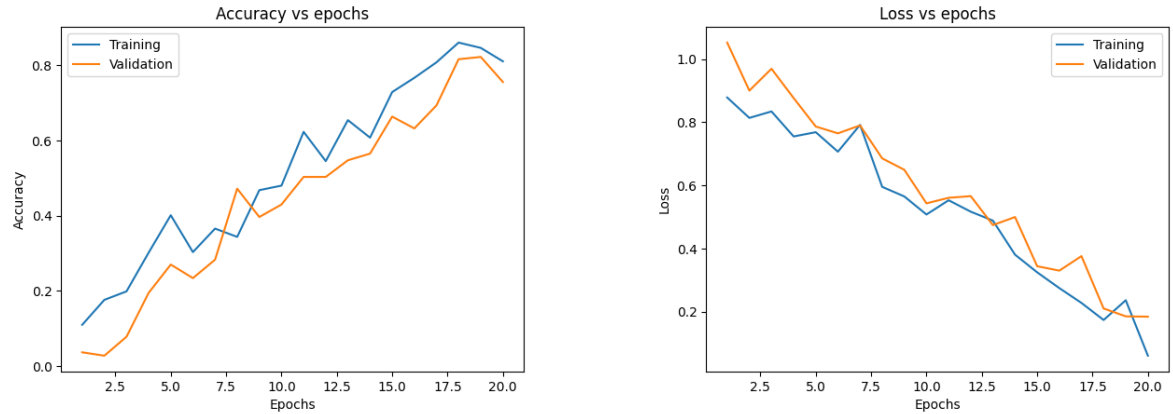


Figure 4.13: CNN Accuracy and Loss Curve

The CNN Accuracy and Loss Curves show the model's performance over time. The accuracy curve, despite some fluctuations, generally trends upward, indicating improved classification. The loss curve, while also variable, trends downward, reflecting reduced prediction errors. These curves demonstrate the model's learning progress.

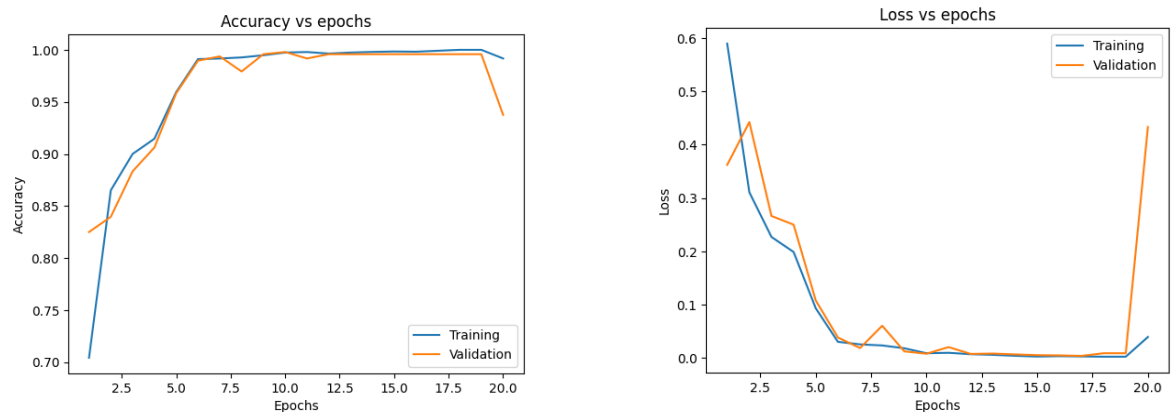


Figure 4.14: Inception V3 Accuracy and Loss

The Inception V3 Accuracy and Loss Curves illustrate the model's performance over time. The accuracy curve, showing a more stable upward trend, indicates consistent improvement

in classification accuracy. The loss curve steadily decreases, reflecting a reduction in prediction errors. These curves highlight the model’s effective learning and better performance compared to the CNN.

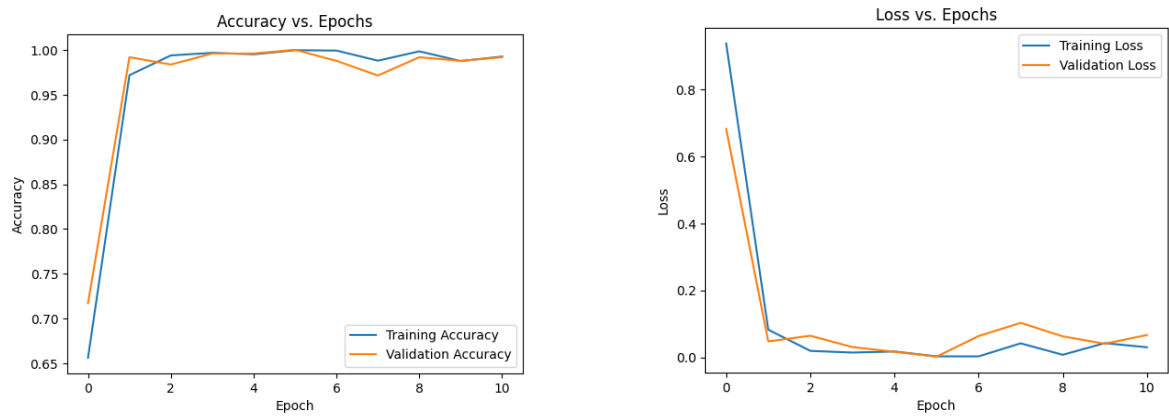


Figure 4.15: ViT Base Accuracy and Loss

The ViT Base Accuracy and Loss Curves demonstrate the model’s performance throughout training. The accuracy curve shows a steady and consistent rise, indicating robust improvement in classification accuracy. Similarly, the loss curve exhibits a continuous decline, suggesting effective reduction of prediction errors.

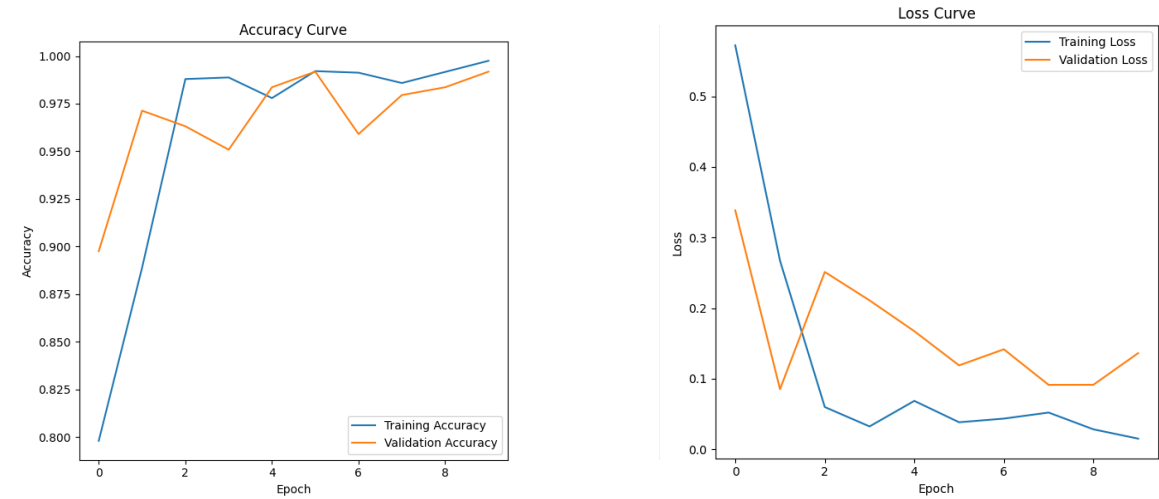


Figure 4.16: DeiT Accuracy and Loss

The DeiT curves depict stable progress with minor fluctuations, showcasing consistent improvement in accuracy and reduced prediction errors.

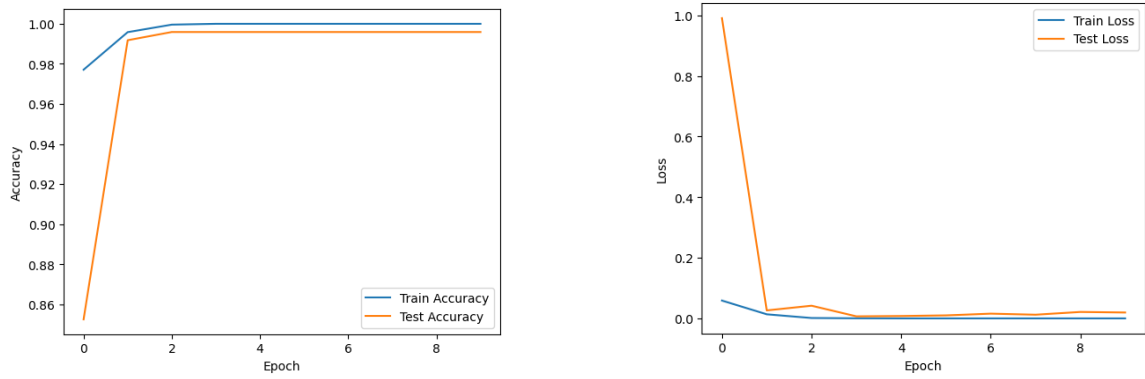


Figure 4.17: BeiT Accuracy and Loss

The BeiT Accuracy and Loss Curves show steady progress during training, with slight fluctuations. The accuracy curve rises consistently, indicating improved performance over epochs. Simultaneously, the loss curve steadily declines, reflecting reduced prediction errors.

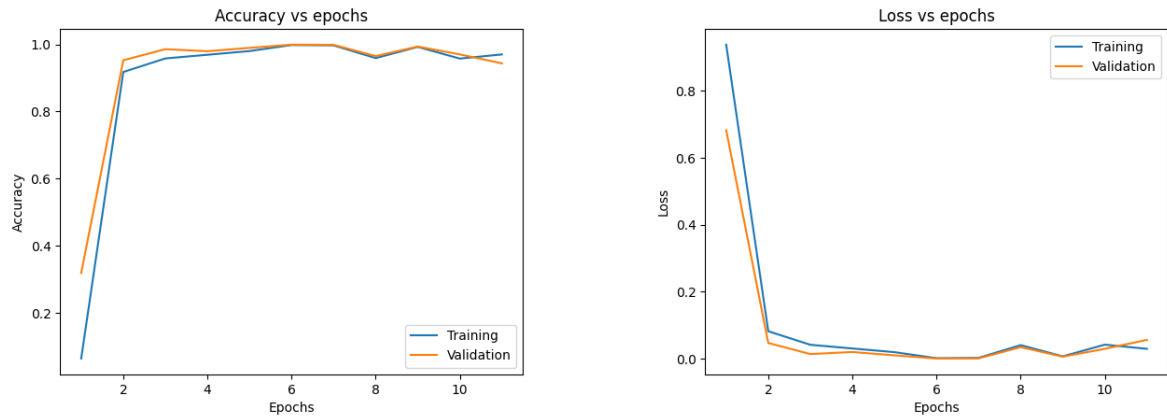


Figure 4.18: ViT Large Accuracy and Loss

ViT Large curves show steady-state improvements, demonstrating the model's ability to cope with data complexity and improve performance over time, demonstrating its robustness and effectiveness in tackling challenging classification challenges.

4.3.1 Conclusion

In conclusion, the accuracy and loss curve for the PCOS detection task provide a visual representation of the model's performance over time. The accuracy curve shows the model's ability to correctly classify examples in the test set, while the loss curve shows the model's ability to minimize the error during the training process

4.4 Comparative Analysis

Approach / Scenario	CNN	Inception v3	PCONet	ViT base	DeiT	BeiT	ViT large
Hossain et al. ^[14]		96.56	98.12				
Our Proposed	95	96		98.68	99	99	99.18

Table 4.1: Comparative Analysis

Metrics	Value
Accuracy	0.991803
Precision	1.000000
Recall	0.995902
F1 Score	0.995885

Table 4.2: Performance Metrics for PCOS Detection

4.5 Experimental Input and Output

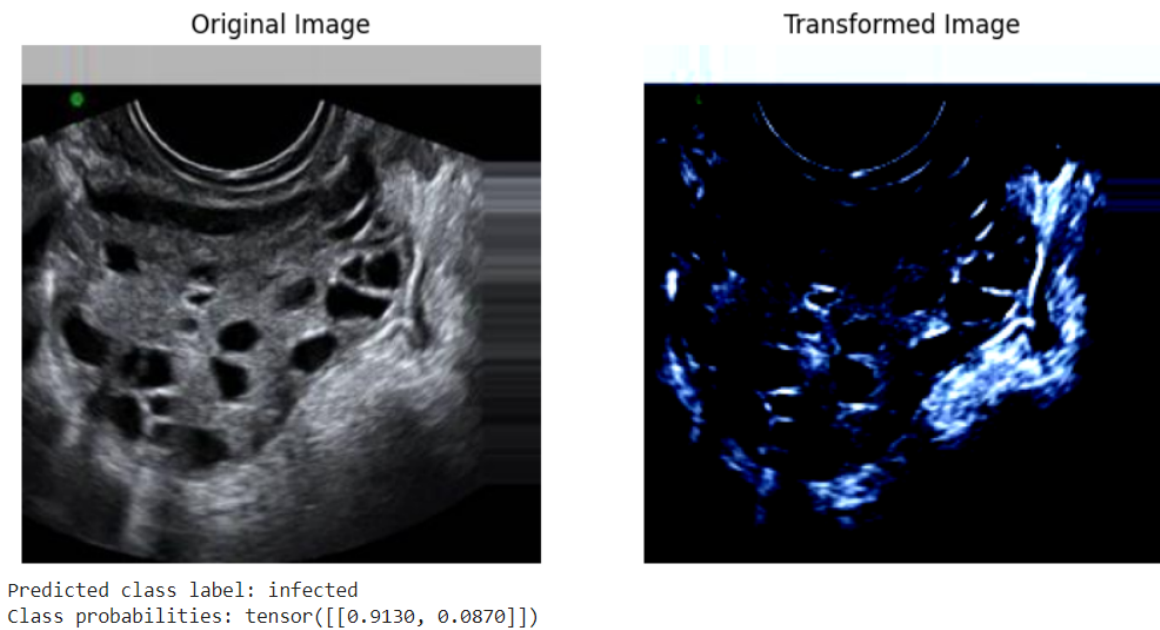


Figure 4.19: Final Outcome

Chapter 5

Conclusion

5.1 Summery of findings

The study showed that the Vision Transformer model with ViT-base and ViT-large is more effective in analyzing medical images. ViT-large and ViT-base both performed well in identifying details in these pictures. With 99.18% accuracy, ViT-large outperforms deep learning models such as Inception V3. These impressive results highlight the ViT model's immense potential to improve applications such as medical image processing and provide promising opportunities for further healthcare research.

5.2 Conclusion

In conclusion, automatic PCOS detection from ultrasound images using Vision Transformer and deep learning is beneficial tool for medical and clinical research. The study found that vision transformer architectures, such as ViT-base, DeiT, BeiT, ViT-large, can be effectively used to identify medical image with high accuracy. Additionally, traditional deep learning models such as CNN, Inception v3 also showed promising results.

5.3 Summery of contribution

Our contribution in this study is to apply binary classification and image preprocessing techniques to improve medical image classification performance on clinical images. The study showed that the Vision Transformer programs, especially the ViT Base, DeiT, BeiT and ViT Large models, are effective in recognizing medical images with high accuracy. The ViT Large model achieved the highest classification accuracy of 99.18%, while the Inception V3 model showed the highest accuracy among traditional deep learning models, with 96.56%.

5.4 Future work

- Increasing data set types and sizes to increase model generalizability and deal with data imbalances.
- Integrating domain-specific medical knowledge, such as ultrasound imaging features, to improve model performance.
- Evaluating model versatility and ensuring robustness and applicability by testing a range of ultrasound and image-based services beyond PCOS detection.

References

- [1] Goodman NF, Cobin RH, Futterweit W, Glueck JS, Legro RS, Carmina E (November 2015). "American Association of Clinical Endocrinologists, American College of Endocrinology, and Androgen Excess and PCOS Society Disease State Clinical Review: Guide to the Best Practices in the Evaluation and Treatment of Polycystic Ovary Syndrome-Part 1". *Endocrine Practice*, 21(11): 1291–1300. doi:10.4158/EP15748.DSC . PMID 26509855.
- [2] I. F. Stein, "Amenorrhea associated with bilateral polycystic ovaries," *Am J Obstet Gynecol*, vol. 29, pp. 181–191, 1935.
- [3] B. Cahyono et al., "An Implementation of Convolutional Neural Network on PCO Classification based on Ultrasound Image," in *Proceedings of the International Conference on Information and Communication Technology (ICoICT)*, 2017.
- [4] L. E. Kenigsberg, C. Agarwal, S. Sin et al., "Clinical utility of magnetic resonance imaging and ultrasonography for diagnosis of polycystic ovary syndrome in adolescent girls," *Fertility and Sterility*, vol. 104, no. 5, pp. 1302–1309, 2015.
- [5] F. S. Abdullah and N. Jondri, "Modified Backpropagation Algorithm for Polycystic Ovary Syndrome Detection Based on Ultrasound Images," in *Proceedings of the International Conference on Soft Computing and Data Mining*, Berlin, Germany, December 2017.
- [6] M. D. Kahsar-Miller, C. Nixon, L. R. Boots, R. C. Go, and R. Azziz, "Prevalence of polycystic ovary syndrome (PCOS) in first-degree relatives of patients with PCOS," *Fertility and Sterility*, vol. 75, no. 1, pp. 53–58, 2001.
- [7] S. Liu, Y. Wang, X. Yang et al., "Deep learning in medical ultrasound analysis: a review," *Engineering*, vol. 5, no. 2, pp. 261–275, 2019.
- [8] M. Sumathi, P. Chitra, R. Sakthi Prabha, and K. Srilatha, "Study and detection of PCOS related diseases using CNN," *IOP Conf. Ser., Mater. Sci. Eng.*, vol. 1070, Nov. 2021, Art. no. 012062.
- [9] S. Kumari, "Classification of PCOS/PCOD using transfer learning and GAN architectures to generate pseudo ultrasound images," Ph.D. dissertation, National College of Ireland, Dublin, 2021.
- [10] B. Cahyono, M. S. Mubarak, and U. N. Wisesty. "An implementation of convolutional neural network on PCO classification based on ultrasound image," in *Proc. 5th Int. Conf. Inf. Commun. Technol. (ICoIC)*, May 2017, pp. 1–4.
- [11] W. Lv, Y. Song, R. Fu, X. Lin, Y. Su, X. Jin, H. Yang, X. Shan, W. Du, Q. Huang et al., "Deep learning algorithm for automated detection of polycystic ovary syndrome using scleral images," *Frontiers in Endocrinology*, vol. 12, 2021.
- [12] R. M. Dewi, Adiwijaya, U. N. Wisesty, and Jondri, "Classification of polycystic ovary based on ultrasound images using competitive neural network," *J. Phys., Conf.*, vol. 971, Mar. 2018, Art. no. 012005.
- [13] S. Rihana, H. Moussallem, C. Skaf, and C. Yaacoub, "Automated algorithm for ovarian cysts detection in ultrasonogram," in *2013 2nd International Conference on Advances in Biomedical Engineering*, IEEE, 2013, pp. 219–222.

- [14] A. K. M. S. Hosain, M. H. K. Mehedi, and I. E. Kabir, "PCONet: A convolutional neural network architecture to detect polycystic ovary syndrome (PCOS) from ovarian ultrasound images," 2022, arXiv:2210.00407.
- [15] P. Soni and S. Vashisht, "Image segmentation for detecting polycystic ovarian disease using deep neural networks," *International Journal of Computer Sciences and Engineering*, vol. 7, no. 3, pp. 534–537, 2019.
- [16] S. Bhosale, L. Joshi, and A. Shivsharan. PCOS (Polycystic Ovarian NDROME) Detection Using Deep Learning. Accessed: May 16, 2022.
- [17] S. Srivastava, P. Kumar, V. Chaudhry, and A. Singh, "Detection of ovarian cyst in ultrasound images using fine-tuned vgg-16 deep learning network," *SN Computer Science*, vol. 1, no. 2, pp. 1–8, 2020.
- [18] A. Alamoudi, I. U. Khan, N. Aslam, N. Alqahtani, H. S. Alsaif, O. Al Dandan, M. Al Gadeeb, and R. Al Bahrani, "A deep learning fusion approach to diagnosis the polycystic ovary syndrome (PCOS)," *Appl. Comput. Intell. Soft Comput.*, vol. 2023, pp. 1–15, Feb. 2023.
- [19] B. Purnama, U. N. Wisesti, F. Nhita, A. Gayatri, and T. Mutiah, "A classification of polycystic ovary syndrome based on follicle detection of ultrasound images," in *Proc. 3rd Int. Conf. Inf. Commun. Technol. (ICoICT)*, May 2015, pp. 396–401.
- [20] S. S. Deshpande and A. Wakankar, "Automated detection of polycystic ovarian syndrome using follicle recognition," in *Proc. IEEE Int. Conf. Adv. Commun., Control Comput. Technol.*, May 2014, pp. 1341–1346.
- [21] L. Thara and T. M. Divya, "Detection and prediction system for polycystic ovary syndrome using structural normalized square similarity detection approach," *Nveo-Natural Volatiles Essential Oils J.*, vol. 8, no. 6, pp. 2834–2842, 2021.
- [22] K. Maheswari, T. Baranidharan, S. Karthik, and T. Sumathi, "Modelling of F3I based feature selection approach for PCOS classification and prediction," *Journal of Ambient Intelligence and Humanized Computing*, vol. 12, no. 1, pp. 1349–1362, 2020.
- [23] C. Gopalakrishnan and M. Iyapparaja, "Detection of polycystic ovary syndrome from ultrasound images using SIFT descriptors," *Bonfring International Journal of Software Engineering and Soft Computing*, vol. 9, no. 2, pp. 26–30, 2019.
- [24] E. Erdemir and Ç. B. ERDAŞ, "Evaluation of Deep Learning Techniques in the Diagnosis of Polycystic Ovary Syndrome," in *2023 Medical Technologies Congress (TIPTEKNO)*, Famagusta, Cyprus, 2023, pp. 1–4.
- [25] J. Patterson and A. Gibson, *Deep Learning: A Practitioner's Approach*, 1st edition, O'Reilly Media, 2017.
- [26] Phung, & Rhee. (2019). A High-Accuracy Model Average Ensemble of Convolutional Neural Networks for Classification of Cloud Image Patches on Small Datasets. *Applied Sciences*, 9, 4500 . doi:10.3390/app9214500.
- [27] Szegedy, Christian, et al. "Rethinking the inception architecture for computer vision." *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016.

- [28] Ali, Luqman and Alnajjar, Fady and Jassmi, Hamad and Gochoo, Munkhjargal and Khan, Wasif and Serhani, Mohamed. (2021). "Performance Evaluation of Deep CNN-Based Crack Detection and Localization Techniques for Concrete Structures." *Sensors*, 21, 1688. doi:10.3390/s21051688.
- [29] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin. "Attention is all you need." *Advances in neural information processing systems*, 30, 2017.
- [30] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby. "An image is worth 16x16 words: transformers for image recognition at scale." *arXiv preprint*, 2021. doi: arXiv:2010.11929v2.
- [31] M. Raghu, T. Unterthiner, S. Kornblith, C. Zhang, and A. Dosovitskiy. "Do vision transformers see like convolutional neural networks?" *Advances in Neural Information Processing Systems*, 34:12116–12128, 2021.
- [32] Y. Lecun, L. Bottou, Y. Bengio and P. Haffner, "Gradient-based learning applied to document recognition," in *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278-2324, Nov. 1998, doi: 10.1109/5.726791.
- [33] Touvron, H., Cord, M., Douze, M., Massa, F., Sablayrolles, A. & Jegou, H.. (2021). Training data-efficient image transformers & distillation through attention. *Proceedings of the 38th International Conference on Machine Learning*, in *Proceedings of Machine Learning Research* 139:10347-10357. Available from <https://proceedings.mlr.press/v139/touvron21a.html>.
- [34] Bao, H., Dong, L., Piao, S., & Wei, F. (2021). Beit: Bert pre-training of image transformers. *arXiv preprint arXiv:2106.08254*.

